Avichal Khanna

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Education

- **Manipal University Jaipur** — Jaipur, India
  Bachelor of Technology - Data Science; CGPA: 9.89 (current) — *July 2024 - June 2028*
  Relevant Coursework: Deep Learning, Data Structures, AI, Machine Learning, Networking, Databases

Summary

B.Tech Data Science student at Manipal University (CGPA 9.89) entering the third year in July, with hands-on experience building production-grade ML systems including LLM inference optimization, RLHF pipelines, and multimodal VLMs. Seeking ML/AI internship at scale.

Projects

- **LLM Inference Optimization Engine | 2026 (MLOps, Systems, LLMs) [GitHub]**: Engineered a high-throughput serving system for LLaMA-2-7B featuring continuous batching, KV-cache management, and INT8 quantization via bitsandbytes. Achieved **4.2x throughput improvement** and **60% memory reduction** over naive serving, benchmarked against vLLM baseline at production scale. Tech: Python, PyTorch, CUDA, Triton Inference Server, Docker, Kubernetes
- **RLHF Fine-tuning Pipeline | 2026 (LLMs, Reinforcement Learning) [GitHub]**: Built an end-to-end RLHF pipeline on GPT-2 covering supervised fine-tuning, reward model training on human preference data, and PPO-based policy optimization via TRL. Achieved **73% reduction in harmful outputs** on a custom safety benchmark vs base model — replicating alignment techniques used in production LLMs. Tech: PyTorch, TRL, HuggingFace, Weights & Biases
- **Sparse MoE Transformer | 2026 (NLP, Systems, Deep Learning) [GitHub]**: Implemented a Sparse Mixture-of-Experts Transformer from scratch in PyTorch, replicating Google Switch Transformer architecture. Trained on WikiText-103 with 8-expert routing, achieving **2.4x throughput over dense baseline** at equivalent perplexity. Engineered load balancing loss to prevent expert collapse and match reported Switch Transformer efficiency gains. Tech: PyTorch, CUDA, Weights & Biases, HuggingFace
- **PRism | 2026 (MLOps, Agentic AI, Full-Stack) [GitHub] | [Live Demo]**: Built and shipped an AI-powered GitHub PR review agent that listens to repository webhook events, runs automated code analysis via Groq (LLaMA-3.3-70B), and surfaces structured review comments with severity tagging. Tracks review history in Supabase and exposes a live React dashboard — reducing manual PR review overhead to zero. Tech: FastAPI, Groq, Supabase, React/Vite, Render
- **YouTube Shorts Automation Pipeline | 2026 (Agentic AI, Systems) [GitHub] | [YouTube]**: Designed and deployed a fully autonomous content pipeline that discovers developer portfolio websites, records them via headless browser, generates contextual scripts using Gemini, synthesizes voiceovers via Groq TTS, burns word-by-word animated subtitles via FFmpeg, and auto-uploads finished Shorts to YouTube — zero human intervention post-trigger. Tech: Python, FFmpeg, Gemini API, Groq TTS, FastAPI, React/Vite
- **Neural Information Retrieval Engine | 2025 (NLP, Search, MLOps) [GitHub]**: Built a production-grade hybrid dense-sparse retrieval system combining BM25 with a fine-tuned bi-encoder (BERT) and FAISS for approximate nearest-neighbour search over a **1M+ document corpus**. Achieved **MRR@10 of 0.38**, outperforming BM25 baseline by 31%, served via FastAPI with **sub-50ms p99 latency**. Tech: PyTorch, FAISS, Elasticsearch, FastAPI, Docker, AWS

Skills Summary

- **Languages**: Python, C++, Java, JavaScript, TypeScript, SQL, Bash
- **ML / AI**: TensorFlow, PyTorch, Keras, Scikit-Learn, HuggingFace Transformers, NLTK, SpaCy, OpenCV
- **Frameworks**: Django, Flask, FastAPI, Node.js, Express.js, REST APIs, GraphQL
- **Cloud & DevOps**: AWS (EC2, S3, Lambda), GCP (BigQuery, Vertex AI), Docker, Kubernetes, CI/CD, Git, GitHub
- **Databases**: PostgreSQL, MySQL, SQLite, Redis
- **Systems**: Linux, Distributed Systems, Multithreading, Raspberry Pi, Arduino
- **Soft Skills**: Cross-functional Leadership, Technical Communication, Agile/Scrum, Problem Solving

Honors and Awards

- **Inspire Award Manak (DST, Govt. of India)**: Awarded **Rs. 10,000 government grant** by the Department of Science & Technology to independently research and build an innovative science project; selected among top district-level nominees nationwide
- **Cosmos Hackathon — 1st Place, Audience Favourite**: Ranked 1st in Audience Favourite category among **400+ participants**; built and pitched a working ML solution under 24 hours
- **Dean's List — 3 Consecutive Semesters**: Recognised in top 1% of Data Science cohort at Manipal University Jaipur; maintained CGPA 9.89/10 across all semesters

Volunteer Experience

- **Joint Secretary at Cyber Space Club** — Jaipur, India
  Leading a team of 150+ members, handling events impacting 15,000+ students — *April 2026 – Present*